```
mimport pandas as pd

# Crop health dataset
data = {
"Yellow_Leaves": [1, 1, 0, 0, 1, 0, 1, 0],
"Spots_Present": [1, 0, 0, 0, 1, 0, 0, 0],
"Disease": ["Fungal", "Nutrient", "Healthy",
"Healthy", "Fungal", "Healthy",
"Nutrient", "Healthy"]
}

df = pd.DataFrame(data)
(df)
```

	Yellow_Leaves	Spots_Present	Disease
0	1	1	Fungal
1	1	0	Nutrient
2	0	0	Healthy
3	0	0	Healthy
4	1	1	Fungal
5	0	0	Healthy
6	1	0	Nutrient
7	0	0	Healthy

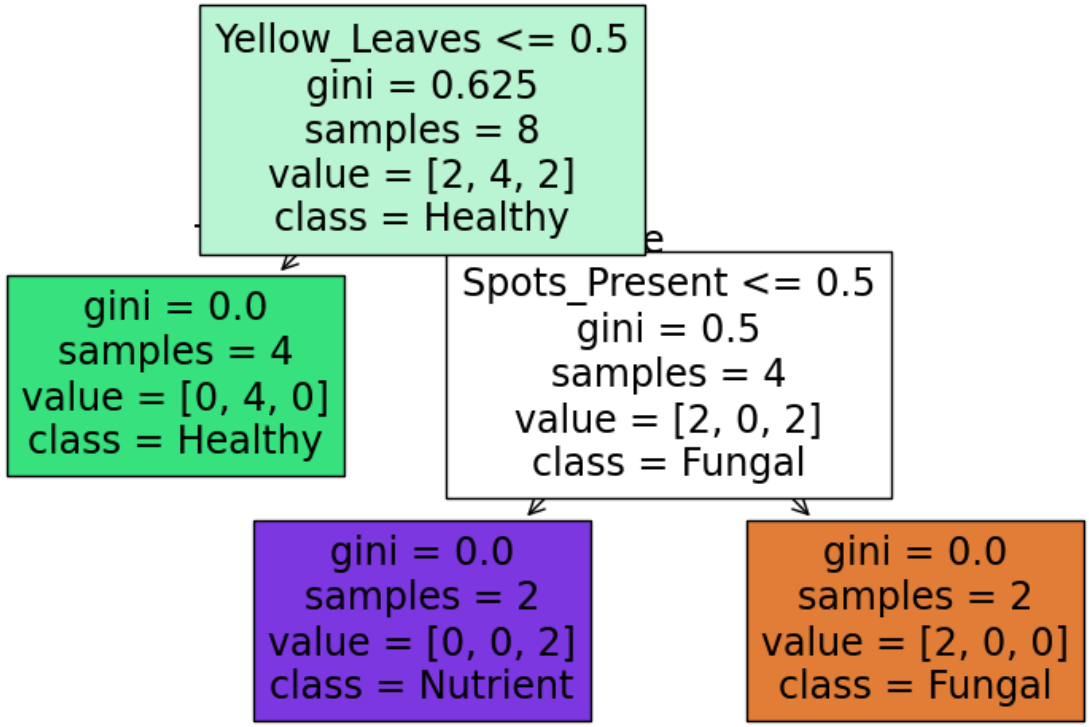
```
from sklearn.tree import DecisionTreeClassifier

X = df[["Yellow_Leaves", "Spots_Present"]] # Symptoms
y = df["Disease"] # Diagnosis

model = DecisionTreeClassifier(random_state=42)
model.fit(X, y)
```

```
from sklearn.tree import plot_tree
import matplotlib.pyplot as plt

plt.figure(figsize=(10,6))
plot_tree(model,
feature_names=X.columns,
class_names=model.classes_,
filled=True)
plt.show()
```




```
import pandas as pd

# Farm dataset with many tracked factors
data = {
"Rainfall_mm": [200, 180, 220, 210, 190, 230],
"Soil_Moisture": [30, 28, 35, 33, 29, 36],
"Humidity": [80, 75, 85, 82, 78, 88],
"Wind_Speed": [10, 12, 9, 11, 10, 8],
"Yield_kg": [3200, 3000, 3500, 3400, 3100, 3600]
}

df = pd.DataFrame(data)
df
```

	Rainfall_mm	Soil_Moisture	Humidity	Wind_Speed	Yield_kg
0	200	30	80	10	3200
1	180	28	75	12	3000
2	220	35	85	9	3500
3	210	33	82	11	3400
4	190	29	78	10	3100
5	230	36	88	8	3600

```
from sklearn.ensemble import RandomForestRegressor

X = df.drop("Yield_kg", axis=1)
y = df["Yield_kg"]
```

```
model = RandomForestRegressor(random_state=42)
model.fit(X, y)
```

```
importances = model.feature_importances_

feature_importance = pd.Series(
importances,
index=X.columns
)
feature_importance
```

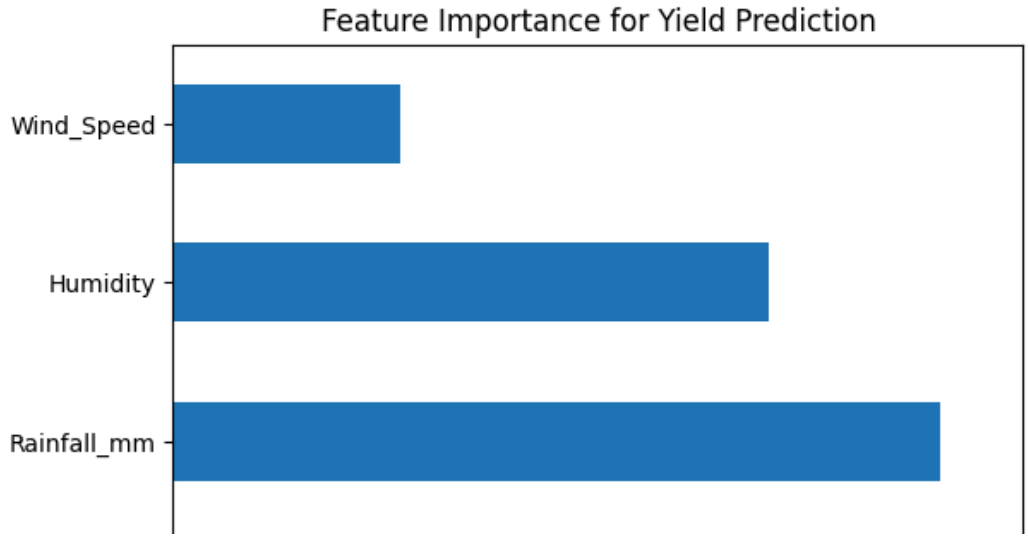
	0
Rainfall_mm	0.319859
Soil_Moisture	0.337573
Humidity	0.248088
Wind_Speed	0.094479

dtype: float64

```
feature_importance.sort_values(ascending=False)
```

	0
Soil_Moisture	0.337573
Rainfall_mm	0.319859
Humidity	0.248088
Wind_Speed	0.094479

dtype: float64

```
feature_importance.sort_values(ascending=False).plot(
    kind="barh",
    title="Feature Importance for Yield Prediction"
)
```

```
<Axes: title={'center': 'Feature Importance for Yield Prediction'}>
```